

To whom it may concern.

São Carlos (Brazil), 7th of April, 2026

Subject: Recommendation letter for MSc. António José Freitas Machado Dórdio Gomes

I am pleased to write this letter of recommendation for António José Freitas Machado Dórdio Gomes, whom I have known for approximately the past two and a half years as a student in the Master's programme in Mechanical Engineering at the Faculty of Engineering of the University of Porto (FEUP). During this period, I had the opportunity to interact with Mr. Gomes both as his professor in courses related to aeronautical structural design and analysis, and as the supervisor of his Master's thesis.

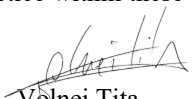
Over this time, he consistently demonstrated strong analytical skills, technical maturity, and a rigorous engineering mindset. His work has been marked by a high level of commitment and intellectual curiosity, as well as the ability to approach complex engineering problems in a structured and methodical way. During the courses Aeronautical Structural Design and Analysis, Mr. Gomes developed extensive work on the structural analysis of an aircraft wing-box, addressing both static and fatigue behaviour. These projects involved the development of detailed finite element models using ABAQUS, the implementation of aerodynamic load cases, and the assessment of the structural response according to certification-oriented design philosophies such as infinite life, safe-life, and damage tolerance approaches. The work required the integration of several aspects of structural mechanics, including composite structural analysis, fatigue assessment, and fracture mechanics, together with careful modelling and interpretation of the numerical results. Throughout these projects, he showed a clear ability to translate theoretical concepts into practical computational models, while maintaining a critical engineering perspective when analyzing the results.

His Master's thesis, entitled "Low-Velocity Impact Damage Assessment and Numerical Modelling of Glass/PPS Laminated Composites Using Commercial Finite Element Tools", further confirmed his ability to carry out research in the field of aerospace composite structures. In this work, Mr. Gomes developed a numerical modelling framework using Abaqus/Explicit to investigate the impact response and damage evolution in thermoplastic composite laminates. The study included the implementation of intralaminar damage modelling using Hashin criteria, cohesive zone modelling for interlaminar delamination, and a bi-phasic modelling strategy, together with a dedicated investigation of the influence of matrix plasticity on the overall impact response. The numerical results were systematically compared with experimental impact data and non-destructive testing measurements, allowing a quantitative assessment of the damage mechanisms involved. One aspect I would particularly highlight in his work is the ability to conduct complex technical studies with a high degree of autonomy. During part of his thesis work carried out abroad he benefited from close technical guidance, but after returning to Portugal he continued the development of the numerical models and analyses largely independently. This transition clearly demonstrated his initiative, perseverance, and ability to critically evaluate modelling strategies and simulation results while maintaining a rigorous and methodical approach to engineering problems.

Beyond his technical capabilities, Mr. Gomes stands out for his initiative, reliability, and ability to communicate technical ideas clearly and effectively. He is a motivated and disciplined engineer who is capable of managing complex technical tasks and delivering well-structured results.

Based on my experience supervising his work over the past years, I am confident that he possesses the technical preparation, intellectual curiosity, and professional attitude required to succeed in any engineering environment, particularly in fields related to aeronautical structures, structural simulation, and composite materials. For these reasons, I recommend him without reservation for opportunities in research, development, or engineering practice within these areas. Please feel free to contact me should you require any further information.

Sincerely yours,



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